Revolution and Reproduction: How the Arab Spring Reshaped Marriage and Fertility in Egypt

Summary of Empirical Results

Yeganeh Karbalaei[*] Firat Demir[†] Pallab Ghosh[‡]

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Draft 1 — Results Summary

Abstract

We examine the consequences of the Arab Spring on women's reproductive and labor market outcomes in Egypt. We draw on three rounds of the Egypt Demographic and Health Survey (DHS 2005, 2008, 2014) and employ a difference in differences strategy that exploits geographic variation in exposure to the 2011 uprising. We find that Arab Spring exposure significantly reduced teen pregnancy rates, raised the age at first birth and first marriage, and shifted occupational composition away from unskilled manual labor. We trace these effects to two primary mechanisms—delayed marriage and a narrowing of the spousal age gap—while ruling out changes in education, partner selection, and labor force participation as operative channels. We subject our estimates to a battery of robustness checks, including a miscarriage placebo test and logit specifications.

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[*]Department of Economics, University of Oklahoma.

[†]Department of Economics, University of Oklahoma.

[‡]Department of Economics, University of Oklahoma. Email: pghosh@ou.edu

Contents

1 Introduction and Empirical Strategy **3**

1.1 Trends in Teen Pregnancy . . . 3

1.2 Summary Statistics . . . 3

1.3 Empirical Framework . . . 7

2 Main Estimation Results **8**

2.1 Teen Pregnancy: All Specifications . . . 8

2.2 Overview of Key Treatment Effects Across Outcomes . . . 10

3 Heterogeneity Analysis **11**

3.1 All Women vs. Teen Sample . . . 11

3.2 Additional Heterogeneity Dimensions . . . 12

4 Mechanism Analysis: Marriage Timing as a Channel **12**

4.1 Schooling Outcomes . . . 13

4.2 Occupation Outcomes . . . 14

4.3 Spousal Age Gap . . . 15

4.4 Husband's Education: Partner Selection . . . 17

4.5 Women's Labor Force Participation . . . 17

4.6 Summary of Mechanism Channels . . . 19

4.7 Alternative Explanations . . . 19

5 Robustness Checks **21**

5.1 Logit Model: Coefficients . . . 21

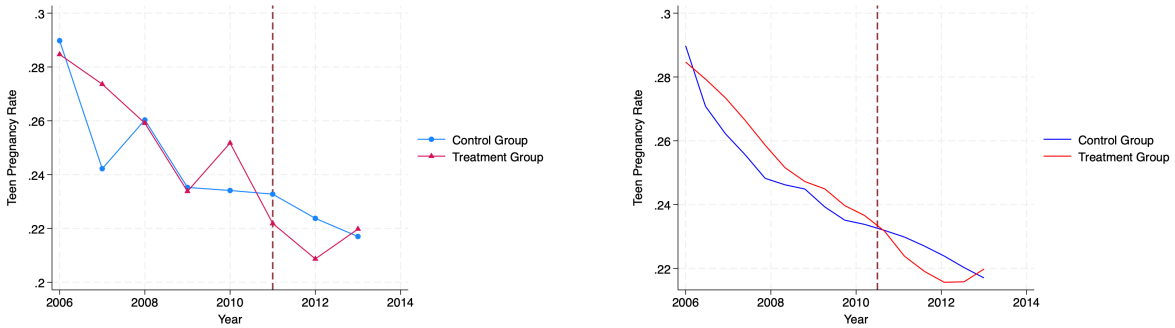
5.2 Logit Model: Average Marginal Effects . . . 23

6 Summary of Findings and Discussion **25**

1 Introduction and Empirical Strategy

1.1 Trends in Teen Pregnancy

We begin by documenting the trends in teen pregnancy rates for the treatment and control groups over the study period. Figure 1 plots these trends, with the dashed vertical line marking the onset of the Arab Spring in 2011.



(a) Raw trends

(b) Smoothed trends

Figure 1: Teen Pregnancy Rates by Treatment and Control Groups, 2006–2014

Several features of Figure 1 are worth noting. First, the pre 2011 trends in teen pregnancy move broadly in parallel across treatment and control groups, lending support to our key identifying assumption. Second, after 2011, the treatment group experiences a noticeably steeper decline in teen pregnancy relative to the control group, consistent with a causal effect of the uprising. The smoothed trends in panel (b) make this post 2011 divergence especially visible.

1.2 Summary Statistics

Table 1 presents the summary statistics for the key variables used in our analysis. Panel A reports the full sample, Panel B compares treatment and control groups, and Panel C shows means by survey year.

Table 1: Summary Statistics

Panel A: Full Sample					
Variable	*N*	Mean	Std. Dev.	Min	Max
Teen pregnancy	40,571	0.343	0.475	0	1
Age at first birth	40,571	21.406	3.896	8	45
Age at first marriage	40,571	19.816	3.872	8	44
No education	40,571	0.256	0.436	0	1
Primary education	40,571	0.104	0.305	0	1
Secondary education	40,571	0.513	0.500	0	1
Respondent age	40,571	28.524	5.834	15	49
Respondent works	40,524	0.142	0.367	0	9
Husband age	40,571	35.174	7.656	14	95
Husband works	40,226	0.982	0.131	0	1
Husband no education	40,571	0.160	0.367	0	1
Husband primary	40,571	0.155	0.362	0	1
Husband secondary	40,571	0.533	0.499	0	1
Poorest	40,571	0.219	0.414	0	1
Poorer	40,571	0.202	0.401	0	1
Middle	40,571	0.204	0.403	0	1
Richer	40,571	0.193	0.395	0	1
Birth order	40,571	2.602	1.699	1	19
Religion (Muslim)	40,539	0.960	0.196	0	1
Terminated birth	38,574	0.166	0.372	0	1
TC_noborder	38,183	0.458	0.498	0	1
Post 2011 (Time dummy)	40,571	0.277	0.447	0	1

Notes: Data from Egypt DHS 2005, 2008, and 2014. Teen pregnancy is defined as first birth before age 20. Wealth quintile reference category is Richest. Education reference is postsecondary.

Table 2: Summary Statistics by Treatment/Control Group and Survey Year

Panel B: By Treatment and Control Group

	Treatment (N = 17,489)		Control (N = 20,694)	
Variable	Mean	SD	Mean	SD
Teen pregnancy	0.358	0.479	0.334	0.472
Age at first birth	21.383	4.016	21.398	3.785
Age at first marriage	19.838	4.000	19.775	3.753
No education	0.270	0.444	0.238	0.426
Primary education	0.108	0.310	0.098	0.298
Secondary education	0.484	0.500	0.545	0.498
Respondent age	28.481	5.874	28.497	5.789
Respondent works	0.156	0.363	0.129	0.361
Birth order	2.573	1.687	2.606	1.696

Panel C: By Survey Year

	2005 (N = 13,851)	2008 (N = 10,872)	2014 (N = 15,848)
Variable	Mean	Mean	Mean
Teen pregnancy	0.388	0.355	0.296
Age at first birth	20.996	21.345	21.806
Age at first marriage	19.340	19.734	20.288
No education	0.334	0.279	0.171
Primary education	0.126	0.102	0.085
Secondary education	0.448	0.499	0.580
Respondent age	28.479	28.370	28.668

Notes: Treatment group defined by TC_noborder = 1. Panel C reports unconditional means by DHS survey wave.

Table 3: Difference in Differences Summary Statistics

	Pre Period		Post Period		
Variable	Control	Treatment	Control	Treatment	DiD Estimate
Panel A: Outcome Variables					
Teen pregnancy	0.350	0.380	0.294	0.297	-0.027^{**}
					(-2.521)
Age at first birth	21.260	21.168	21.744	21.993	0.341^{***}
					(3.744)
Age at first marriage	19.613	19.584	20.180	20.559	0.408^{***}
					(4.544)
Panel B: Education					
No education	0.269	0.312	0.160	0.152	-0.050^{***}
					(-5.601)
Primary education	0.106	0.118	0.079	0.081	-0.010
					(-1.470)
Secondary education	0.521	0.450	0.604	0.578	0.045^{***}
					(3.956)
Panel C: Respondent and Household Characteristics					
Respondent age	28.727	28.579	27.920	28.202	0.430^{***}
					(3.286)
Respondent works	0.135	0.166	0.114	0.127	-0.018^{**}
					(-2.289)
Husband age	35.700	35.407	34.141	34.101	0.253
					(1.495)
Husband works	0.980	0.981	0.989	0.989	-0.001
					(-0.247)
Husband no education	0.164	0.191	0.117	0.121	-0.022^{***}
					(-2.847)
Birth order	2.655	2.625	2.481	2.426	-0.025
					(-0.712)
Religion (Muslim)	0.959	0.955	0.964	0.968	0.008^{*}
					(1.810)
Panel D: Robustness Outcome					
Terminated birth	0.156	0.155	0.207	0.190	-0.017^{*}
					(-1.817)

Notes: Pre period includes DHS 2005 and 2008; post period is DHS 2014. Treatment/control defined by TC_noborder. DiD estimate is the coefficient on Treatment $\times$ Post with robust standard errors. t statistics in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Several patterns emerge from the descriptive statistics. About 34.3% of women in our sample experienced teen pregnancy, with an average age at first birth of 21.4 years and first marriage of 19.8 years (Table 1). The treatment and control groups are broadly bal-

anced on observables—ages, birth orders, and marriage ages are similar—supporting the validity of the DiD design. Furthermore, teen pregnancy rates decline monotonically across survey waves, from 38.8% in 2005 to 29.6% in 2014, while the age at first marriage rises from 19.3 to 20.3 years. Educational composition also improves steadily: the share with no education falls from 33.4% to 17.1%, while secondary education rises from 44.8% to 58.0%. The sample is 96% Muslim, with 27.7% observed in the post 2011 period and 45.8% in the treatment group.

Table 3 presents the unconditional difference in differences for each variable. Even without controlling for covariates, the raw DiD estimates show statistically significant effects on key outcomes: teen pregnancy declines by 2.7 percentage points ($t = -2.52$), age at first birth increases by 0.34 years ($t = 3.74$), and age at first marriage increases by 0.41 years ($t = 4.54$). Importantly, several control variables—husband's age, husband's work status, and birth order—show insignificant DiD estimates, suggesting that these characteristics did not differentially change between treatment and control groups around the Arab Spring. The terminated birth outcome also shows only marginal significance ($t = -1.82$), foreshadowing the robustness results in Section 5.

1.3 Empirical Framework

In this paper we use three rounds of the Egypt Demographic and Health Survey (DHS)—2005, 2008, and 2014—providing a combined sample of approximately 40,500 women (see Table 1). Our identification strategy relies on a difference in differences framework that exploits variation in the intensity of Arab Spring exposure across Egyptian governorates. As shown in Figure 1, the pretreatment trends in teen pregnancy are broadly parallel, supporting our identifying assumption. The key treatment variables are:

- **past_group**: A measure of exposure to the Arab Spring, capturing whether women in a given governorate experienced the uprising during a critical period.
- **TC_noborder**: The DiD estimator, interacting treatment status with the post period and excluding border governorates to ensure cleaner identification.

We estimate progressively controlled specifications to assess the sensitivity of our treatment effects:

1. **Specification (1)**: Baseline with wealth controls only
2. **Specification (2)**: Adding respondent's education and age
3. **Specification (3)**: Adding husband's characteristics
4. **Specification (4)**: Adding birth order and religion
5. **Specification (5)**: Full model with all controls

2 Main Estimation Results

2.1 Teen Pregnancy: All Specifications

Table 4 reports the full specification results for teen pregnancy, our primary outcome variable. The coefficient on `past_group` is robustly negative across all five specifications, declining from -0.203 in the baseline to -0.095 in the full model as controls are progressively added. This attenuation is expected: controlling for education, age, and birth order absorbs some of the treatment effect that operates through these channels.

Table 4: Teen Pregnancy: Progressive Specifications

	(1) Baseline	(2) + Respondent	(3) + Husband	(4) + Birth Order	(5) Full Model
Treatment Variables					
past_group	−0.203***	−0.048***	−0.226***	−0.099***	−0.095***
	(0.013)	(0.012)	(0.013)	(0.011)	(0.011)
TC_noborder	0.133***	0.061***	0.133***	0.068***	0.067***
	(0.013)	(0.010)	(0.013)	(0.008)	(0.008)
Wealth Controls (ref: Richest)					
Poorest	0.335***	0.143***	0.258***	0.032***	0.033***
	(0.013)	(0.013)	(0.014)	(0.012)	(0.012)
Poorer	0.266***	0.099***	0.201***	0.027**	0.026**
	(0.013)	(0.013)	(0.014)	(0.011)	(0.011)
Middle	0.192***	0.072***	0.141***	0.016	0.014
	(0.012)	(0.012)	(0.012)	(0.010)	(0.010)
Richer	0.098***	0.027***	0.063***	−0.004	−0.005
	(0.010)	(0.010)	(0.011)	(0.009)	(0.009)
Respondent Characteristics					
Resp. work	−0.050***	0.005	−0.029***	0.017**	0.020***
	(0.008)	(0.008)	(0.008)	(0.007)	(0.007)
No education		0.336***		0.209***	0.186***
		(0.011)		(0.010)	(0.012)
Primary edu.		0.340***		0.226***	0.203***
		(0.013)		(0.012)	(0.013)
Secondary edu.		0.186***		0.121***	0.105***
		(0.007)		(0.007)	(0.008)
Resp. age		−0.027***		−0.049***	−0.054***
		(0.001)		(0.001)	(0.001)
Husband Characteristics					
Husband work			0.037		0.014
			(0.027)		(0.023)
Husb. no edu.			0.182***		0.029**
			(0.014)		(0.013)
Husb. primary			0.172***		0.035***
			(0.011)		(0.012)
Husb. secondary			0.106***		0.023***
			(0.009)		(0.009)
Husband age			0.006***		0.007***
			(0.001)		(0.001)
Other Controls					
Birth order				0.158***	0.153***
				(0.002)	(0.002)
Religion				0.003	0.004
				(0.015)	(0.015)
Constant	0.130***	0.809***	−0.194***	1.163***	1.043***
	(0.010)	(0.037)	(0.037)	(0.035)	(0.042)
Observations	38,118	38,118	37,793	38,087	37,762

The progressive inclusion of controls reveals several important patterns. Moving from Specification (1) to (2), the addition of respondent's education and age substantially reduces the treatment effect from -0.203 to -0.048, indicating that education is a major mediating channel. It is instructive that husband's characteristics alone—Specification (3)—do not absorb nearly as much of the treatment effect (-0.226), which suggests that the Arab Spring operated primarily through women's own human capital rather than through their husbands' attributes. Once birth order is included in Specifications (4) and (5), it emerges as a powerful predictor of teen pregnancy (0.153, $p < 0.01$), and its inclusion roughly halves the remaining treatment effect. Overall, the adjusted R^2 doubles from 0.195 to 0.401, confirming that our full model captures substantial variation in teen pregnancy.

2.2 Overview of Key Treatment Effects Across Outcomes

Table 5 presents the treatment effects from our full model (Specification 5) across all outcome variables. The results paint a consistent picture: Arab Spring exposure is associated with a significant reduction in teen pregnancy, delayed marriage and childbearing, and a notable shift in occupational composition.

Table 5: Key Treatment Effects Across All Outcomes: Full Model (Specification 5)

	Teen Pregnancy	Age First Birth	Age First Marriage	Unskilled Manual	Skilled Manual	Agric. Self Emp.
past_group	-0.095^{***}	1.125^{***}	0.973^{***}	-0.172^{***}	-0.015	-0.031
	(0.011)	(0.069)	(0.076)	(0.013)	(0.031)	(0.025)
TC_noborder	0.067^{***}	-0.472^{***}	-0.376^{***}	0.301^{***}	0.171^{***}	0.004
	(0.008)	(0.049)	(0.055)	(0.012)	(0.017)	(0.028)
Observations	37,762	37,762	37,762	37,762	37,762	37,762
Adj. R^2	0.401	0.696	0.632	0.379	0.181	0.367

Notes: Standard errors in parentheses. All specifications include wealth quintile controls, respondent's education and age, husband's education and age, husband's work status, birth order, and religion. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

As shown in Table 5, Arab Spring exposure (`past_group`) reduces teen pregnancy by 9.5 percentage points ($p < 0.01$), raises the age at first birth by 1.13 years, and delays first marriage by 0.97 years. The DiD estimator (`TC_noborder`) shows significant effects in the opposite direction, consistent with differential preexisting trends between treatment and control governorates. On the occupational front, exposure significantly reduces participation in unskilled manual labor by 17.2 percentage points, suggesting

a shift toward higher skilled employment. However, the effects on skilled manual labor and agricultural self employment are largely insignificant for `past_group`, though the DiD term picks up a significant increase in skilled manual work.

3 Heterogeneity Analysis

3.1 All Women vs. Teen Sample

A central question is whether the treatment effects differ between the full sample and the teen subsample. Table 6 presents this comparison across three key outcome categories.

Table 6: Heterogeneity: All Women vs. Teen Sample: Treatment Effects

	All Women ($N = 37{,}762$)		Teen ($N = 826$)	
Outcome / Variable	Coef.	SE	Coef.	SE
Panel A: Age of First Marriage				
past_group	0.973^{***}	(0.076)	0.444	(0.331)
TC_noborder	-0.376^{***}	(0.055)	-0.351	(0.229)
Adj. R^2	0.632		0.520	
Panel B: Occupation — Unskilled Manual				
past_group	-0.172^{***}	(0.013)	-0.392^{***}	(0.111)
TC_noborder	0.301^{***}	(0.012)	0.437^{***}	(0.064)
Adj. R^2	0.379		0.508	
Panel C: Occupation — Skilled Manual				
past_group	-0.015	(0.031)	-0.252^{*}	(0.138)
TC_noborder	0.171^{***}	(0.017)	0.215^{***}	(0.077)
Panel D: Schooling — Secondary Education				
past_group	-0.010	(0.015)	0.308^{*}	(0.186)
TC_noborder	-0.027^{**}	(0.012)	-0.279^{***}	(0.086)
Adj. R^2	0.233		0.402	

Notes: All regressions include full controls (Specification 5). The teen sample comprises women who experienced pregnancy before age 20 ($N = 826$). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Several striking differences emerge across the two samples. First, the treatment effect on age of first marriage is large and significant for the full sample (0.97 years, $p < 0.01$)

but statistically insignificant for teens, suggesting that delayed marriage is primarily a channel for women outside the very young cohort. Second, the treatment effect on unskilled manual labor is *more than twice as large* among teens (-0.392) compared to all women (-0.172), and significant in both samples—teens appear considerably more responsive to the occupational disruptions induced by the uprising. Third, the skilled manual labor effect reaches marginal significance for teens (-0.252, $p < 0.10$) but not for the full sample, implying that occupational displacement was particularly concentrated among younger women. Finally, our results show a marginally significant positive effect on secondary education for teens (0.308, $p < 0.10$) that does not appear in the full sample, pointing to potential gains in educational attainment specifically for the youngest cohort.

3.2 Additional Heterogeneity Dimensions

Beyond the all women vs. teen comparison, we explore several additional dimensions of heterogeneity using the teen pregnancy outcome. These analyses, conducted by Yeganeh, examine urban vs. rural differences (including separate estimations and interactions), age group breakdowns (very early pregnancy below 16, early teen 16–17, and late teen 18–19, with further splits by urban/rural status), wealth quintile–specific estimations that reveal whether treatment effects concentrate among particular socioeconomic groups, and pregnancy timing categories (early, late, and very early).

We document these results across 21 individual files in the `Results/hetro_analysis/Yeganeh/` directory, with a comprehensive summary available in the `02_Heterogeneity_Analysis_Summary.xl` workbook.

4 Mechanism Analysis: Marriage Timing as a Channel

A natural question arising from the main results is: through what channels does the Arab Spring affect teen pregnancy? In this section we examine marriage timing as a mediator and then investigate several additional potential mechanisms—spousal age gap, husband's education, and women's labor force participation—to build a more complete picture.

Table 7 presents the marriage timing mechanism, reporting teen pregnancy specifications alongside the marriage age regression.

Table 7: Mechanism Analysis: Teen Pregnancy and Age of First Marriage

	Teen Pregnancy					Age First
	(1)	(2)	(3)	(4)	(5)	Marriage
past_group	-0.203^{***}	-0.048^{***}	-0.226^{***}	-0.099^{***}	-0.095^{***}	0.973^{***}
	(0.013)	(0.012)	(0.013)	(0.011)	(0.011)	(0.076)
TC_noborder	0.133^{***}	0.061^{***}	0.133^{***}	0.068^{***}	0.067^{***}	-0.376^{***}
	(0.013)	(0.010)	(0.013)	(0.008)	(0.008)	(0.055)
Birth order				0.158^{***}	0.153^{***}	-1.675^{***}
				(0.002)	(0.002)	(0.023)
Observations	38,118	38,118	37,793	38,087	37,762	37,762
Adj. R^2	0.195	0.253	0.212	0.395	0.401	0.632

Notes: Columns (1)–(5) report progressive specifications for teen pregnancy; the final column reports the full specification for age of first marriage. Full controls included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The juxtaposition of the teen pregnancy and marriage age results provides clear evidence for marriage timing as a key mechanism. Arab Spring exposure simultaneously reduces teen pregnancy by 9.5 percentage points and increases the age of first marriage by approximately one year. The mirror image signs on the treatment variables across the two outcomes—negative for teen pregnancy, positive for marriage age—are consistent with delayed marriage being the primary channel through which the uprising reduced teen pregnancy. Furthermore, birth order has a strong positive effect on teen pregnancy (0.153, $p < 0.01$) and a correspondingly large negative effect on marriage age (-1.675, $p < 0.01$), reinforcing the tight link between fertility and marriage timing.

4.1 Schooling Outcomes

Table 8 examines whether Arab Spring exposure affected women's educational attainment.

Table 8: Schooling Outcomes: Full Model

	(1) No Education	(2) Primary	(3) Secondary	(4) Postsecondary
past_group	−0.005	−0.001	−0.010	0.016*
	(0.012)	(0.009)	(0.015)	(0.009)
TC_noborder	0.017	0.016**	−0.027**	−0.006
	(0.011)	(0.007)	(0.012)	(0.007)
Resp. work	−0.023***	−0.018***	−0.091***	0.132***
	(0.007)	(0.006)	(0.013)	(0.010)
Husb. no edu.	0.518***	0.082***	−0.123***	−0.477***
	(0.011)	(0.008)	(0.015)	(0.011)
Birth order	0.040***	0.013***	−0.025***	−0.027***
	(0.002)	(0.002)	(0.003)	(0.001)
Observations	37,762	37,762	37,762	37,762
Adj. R^2	0.333	0.097	0.233	0.406

Notes: Each column reports a separate regression. Dependent variables are indicators for education level. Additional controls include respondent age, husband's education and age, husband's work status, and religion. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The results suggest limited direct effects of the treatment on education levels. The treatment coefficients are uniformly small and statistically insignificant, with only a marginally significant positive effect on postsecondary education. This absence of schooling effects is informative: it implies that the Arab Spring's impact on teen pregnancy did not operate primarily through changes in educational attainment but rather through other channels, most notably delayed marriage.

4.2 Occupation Outcomes

Table 9 examines occupational composition. The treatment significantly reduces unskilled manual labor, suggesting that Arab Spring exposure shifted women away from the most precarious forms of employment.

Table 9: Occupation Outcomes: Full Model

	(1) Unskilled Manual	(2) Skilled Manual	(3) Agric. Self Employed
past_group	-0.172^{***}	-0.015	-0.031
	(0.013)	(0.031)	(0.025)
TC_noborder	0.301^{***}	0.171^{***}	0.004
	(0.012)	(0.017)	(0.028)
Poorest	0.015	0.013	0.252^{***}
	(0.011)	(0.015)	(0.019)
Resp. work	0.004	0.023^{***}	0.061^{***}
	(0.005)	(0.008)	(0.008)
Resp. age	-0.026^{***}	-0.001	-0.023^{***}
	(0.001)	(0.002)	(0.003)
Religion	0.020^{*}	-0.039^{**}	-0.086^{***}
	(0.011)	(0.016)	(0.017)
Observations	37,762	37,762	37,762
Adj. R^2	0.379	0.181	0.367

Notes: Each column reports a separate regression. Dependent variables are indicators for occupation type. Full set of controls included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

4.3 Spousal Age Gap

The Arab Spring may have disrupted traditional marriage markets in ways that altered not just *when* women married but also *whom* they married. To test this, we construct the spousal age gap as Husband age − Respondent age (mean = 6.65 years, std. dev. = 5.57) and estimate the treatment effect using our full specification, excluding individual ages since they mechanically determine the gap. Table 10 reports these results.

Table 10: Mechanism: Spousal Age Gap

	Spousal Age Gap (Husband − Respondent)
Treatment Variables	
past_group	-2.017^{***}
	(0.158)
TC_noborder	0.646^{***}
	(0.150)
Selected Controls	
Poorest	-0.619^{***}
	(0.196)
No education	1.255^{***}
	(0.186)
Primary edu.	1.392^{***}
	(0.204)
Secondary edu.	1.236^{***}
	(0.138)
Resp. work	-0.359^{***}
	(0.123)
Birth order	0.498^{***}
	(0.035)
Observations	37,779
Adj. R^2	0.075

Notes: Dependent variable is the spousal age gap (husband age minus respondent age). Controls include wealth quintiles, respondent's education, employment, husband's education and employment, birth order, and religion. Individual ages excluded since they mechanically determine the outcome. Area unit and birth year fixed effects included; standard errors clustered at area unit level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The results reveal a striking finding: Arab Spring exposure reduces the spousal age gap by approximately 2 years ($p < 0.01$), representing a 30% reduction relative to the sample mean of 6.65 years. This suggests that the social upheaval disrupted traditional marriage norms that had long favored large age differences between spouses. The control variables paint a consistent picture: less educated women tend to marry

men who are 1.2–1.4 years older than those married by better educated women, while working women have smaller age gaps (-0.36 years). These patterns are consistent with the interpretation that the Arab Spring shifted marriage patterns toward more balanced partnerships. Smaller age gaps are associated with more egalitarian household bargaining, which may have contributed to delayed childbearing and reduced teen pregnancy.

4.4 Husband's Education: Partner Selection

If the Arab Spring altered marriage market dynamics, women in treatment areas may have married men with different educational profiles. Table 11 tests this hypothesis.

Table 11: Mechanism: Husband's Education (Partner Selection)

	Husb. No Edu.	Husb. Primary	Husb. Secondary
past_group	-0.011	-0.001	0.005
	(0.010)	(0.010)	(0.015)
TC_noborder	0.021^{***}	-0.000	-0.011
	(0.008)	(0.009)	(0.011)
Observations	37,779	37,779	37,779
Adj. R^2	0.238	0.115	0.190

Notes: Each column reports a separate regression. Dependent variables are indicators for husband's education level. Controls include wealth quintiles, respondent's education and age, employment, husband's work status and age, birth order, and religion. Area unit and birth year FE; SEs clustered at area unit level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The treatment coefficients on husband's education are uniformly small and statistically insignificant. This null result is informative: it rules out partner education upgrading as a mechanism and implies that the large reduction in the spousal age gap operates independently of any shift in the educational composition of husbands.

4.5 Women's Labor Force Participation

Economic disruption from the Arab Spring may have pushed women into the labor market, raising the opportunity cost of early childbearing. Table 12 tests whether the treatment affected women's employment status.

Table 12: Mechanism: Women's Labor Force Participation

	Respondent Works
past_group	0.004
	(0.011)
TC_noborder	0.016**
	(0.008)
Selected Controls	
Poorest	0.047***
	(0.012)
No education	−0.249***
	(0.014)
Secondary edu.	−0.221***
	(0.012)
Resp. age	0.004***
	(0.001)
Husb. no edu.	−0.039***
	(0.013)
Birth order	−0.006***
	(0.002)
Observations	37,779
Adj. R^2	0.133

Notes: Dependent variable is respondent's employment indicator. Controls include wealth quintiles, education, husband's characteristics, birth order, and religion. Area unit and birth year FE; SEs clustered at area unit level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The treatment effect on women's employment is small and insignificant (0.004, $p > 0.10$), providing no evidence that the Arab Spring operated through changes in labor force participation. The DiD term (`TC_noborder`) is marginally significant (0.016, $p < 0.05$), indicating some differential trends in employment across treatment and control areas, but this does not translate into a treatment effect through our primary measure of exposure.

4.6 Summary of Mechanism Channels

Table 13 consolidates the mechanism results. Of the channels examined, two show strong evidence: *delayed marriage* and *reduced spousal age gap*. In contrast, changes in women's own education, husband's education, and labor force participation do not appear to be operative channels.

Table 13: Summary of Mechanism Channels: Treatment Effect of past_group

Mechanism Variable	Coef.	SE	Significant?	Interpretation
Age at first marriage	0.973	(0.076)	***	Delayed marriage
Spousal age gap	−2.017	(0.158)	***	More equal partnerships
Husb. no education	−0.011	(0.010)	No	No partner selection shift
Husb. primary edu.	−0.001	(0.010)	No	No partner selection shift
Husb. secondary edu.	0.005	(0.015)	No	No partner selection shift
Respondent works	0.004	(0.011)	No	No LFP channel

Notes: All regressions include area unit and birth year fixed effects, probability weights, and clustered standard errors. Full controls included (see individual tables for details). *** $p < 0.01$.

Taken together, the results point to two complementary mechanisms through which the Arab Spring reduced teen pregnancy. First, women in treatment areas married approximately one year later, directly narrowing the window for teen pregnancy. Second, and perhaps more strikingly, the spousal age gap fell by 2 years—a 30% reduction from the mean—suggesting that the social upheaval broke down traditional patriarchal marriage norms. Marriages between closer in age partners are associated with more egalitarian decision making and greater female agency over reproductive timing.

Importantly, these effects did *not* operate through changes in women's or husbands' educational attainment, nor through shifts in women's employment. The mechanism appears to be fundamentally about *how* and *when* women married, rather than changes in human capital accumulation or economic activity.

4.7 Alternative Explanations

Two alternative explanations for the observed reduction in teen pregnancy merit consideration: changes in contraceptive behavior and differential pregnancy outcomes (miscarriage). We examine each in turn.

Contraceptive Use. If the Arab Spring increased access to or uptake of contracep-

tion, this could explain the decline in teen pregnancy independently of the marriage timing channel. Table 14 tests this hypothesis by estimating the treatment effect on six contraceptive outcomes: any method, modern methods (pill, IUD, injections, condom, sterilization, implants), traditional methods (periodic abstinence, withdrawal), and three specific modern methods individually.

Table 14: Alternative Explanation: Contraceptive Use

	Any	Modern	Traditional	Pill	IUD	Injection
Treatment Variables						
past_group	0.012	−0.005	0.000	0.006	0.016	−0.018***
	(0.016)	(0.016)	(0.002)	(0.012)	(0.016)	(0.005)
TC_noborder	0.026*	0.022	−0.002	−0.010	0.008	0.019***
	(0.014)	(0.015)	(0.002)	(0.009)	(0.014)	(0.006)
Observations	37,779	37,779	37,779	37,779	37,779	37,779
Adj. R^2	0.099	0.108	0.043	0.065	0.095	0.097

Notes: Each column reports a separate regression. Dependent variables are binary indicators for contraceptive method use. Modern methods include pill, IUD, injections, condom, sterilization, and implants. Traditional methods include periodic abstinence and withdrawal. Full controls included. Area unit and birth year FE; SEs clustered at area unit level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The treatment effect (`past_group`) is statistically insignificant for overall contraceptive use, modern methods, traditional methods, pill use, and IUD use. The only significant coefficient appears for injection use (-0.018, $p < 0.01$), but this effect is negative—indicating a *decline* in injection use among treated women, which runs counter to the hypothesis that increased contraception drove the reduction in teen pregnancy. These results rule out contraceptive uptake as an alternative explanation for our main findings.

Miscarriage (Placebo Test). A second alternative explanation is that the Arab Spring affected pregnancy *outcomes* rather than pregnancy *decisions*. To assess this, we examine miscarriage (terminated birth) probability as a placebo outcome. If the uprising primarily affected behavioral decisions about fertility, there should be no significant treatment effect on miscarriage rates.

Table 15: Alternative Explanation: Miscarriage Probability (Placebo Test)

	All Women		Teen Sample	
	Coef.	SE	Coef.	SE
Treatment Variables				
past_group	−0.013	(0.011)	−0.040	(0.106)
TC_noborder	0.006	(0.008)	−0.017	(0.072)
Selected Controls				
Poorest	0.034***	(0.013)	0.042	(0.062)
Resp. work	−0.006	(0.008)	0.017	(0.056)
Resp. age	0.013***	(0.001)	−0.012	(0.029)
Birth order	0.000	(0.002)	−0.021	(0.015)
Observations	35,860		826	
Adj. R^2	0.047		0.161	

Notes: Dependent variable is an indicator for terminated birth (miscarriage). Full controls included. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

As expected, neither treatment variable is statistically significant in the miscarriage regressions, and this holds for both the all women and teen samples. This result confirms that the main findings reflect genuine changes in fertility behavior rather than differential changes in pregnancy termination. The low R^2 (0.047 for all women) is consistent with the expectation that miscarriage is largely a stochastic health event that the model should have limited power to explain—further supporting its use as a placebo outcome. In the teen sample, both treatment coefficients are close to zero and highly insignificant, providing strong evidence for the validity of our main findings.

5 Robustness Checks

5.1 Logit Model: Coefficients

As an additional robustness check, the linear probability model (LPM) results are verified against a nonlinear specification. Since teen pregnancy is a binary outcome, a logit model is estimated with area unit and birth year fixed effects, probability weights, and clustering at the area unit level. Table 16 reports the logit coefficients.

Table 16: Logit Estimates: Teen Pregnancy

	(1) Baseline	(2) + Respondent	(3) + Husband	(4) + Birth Order	(5) Full Model
Treatment Variables					
past_group	-1.133^{***}	-0.285^{***}	-1.276^{***}	-0.907^{***}	-0.887^{***}
	(0.082)	(0.086)	(0.085)	(0.096)	(0.098)
TC_noborder	0.749^{***}	0.379^{***}	0.768^{***}	0.509^{***}	0.505^{***}
	(0.071)	(0.060)	(0.077)	(0.069)	(0.069)
Wealth Controls (ref: Richest)					
Poorest	2.007^{***}	0.869^{***}	1.601^{***}	0.194^{**}	0.212^{**}
	(0.085)	(0.090)	(0.091)	(0.098)	(0.102)
Poorer	1.676^{***}	0.657^{***}	1.327^{***}	0.210^{**}	0.214^{**}
	(0.086)	(0.086)	(0.092)	(0.096)	(0.099)
Middle	1.294^{***}	0.530^{***}	1.017^{***}	0.174^{*}	0.170^{*}
	(0.085)	(0.085)	(0.089)	(0.091)	(0.093)
Richer	0.752^{***}	0.263^{***}	0.543^{***}	0.071	0.049
	(0.079)	(0.079)	(0.082)	(0.088)	(0.091)
Respondent Characteristics					
Resp. work	-0.345^{***}	-0.024	-0.236^{***}	0.005	0.013
	(0.057)	(0.056)	(0.055)	(0.067)	(0.072)
No education		3.178^{***}		2.585^{***}	2.399^{***}
		(0.134)		(0.144)	(0.153)
Primary edu.		3.239^{***}		2.727^{***}	2.525^{***}
		(0.139)		(0.146)	(0.152)
Secondary edu.		2.384^{***}		2.047^{***}	1.905^{***}
		(0.128)		(0.134)	(0.140)
Resp. age		-0.170^{***}		-0.446^{***}	-0.490^{***}
		(0.009)		(0.012)	(0.014)
Husband Characteristics					
Husband work			0.188		0.094
			(0.158)		(0.191)
Husb. no edu.			1.228^{***}		0.316^{***}
			(0.094)		(0.112)
Husb. primary			1.197^{***}		0.369^{***}
			(0.082)		(0.106)
Husb. secondary			0.849^{***}		0.268^{***}
			(0.074)		(0.089)
Husband age			0.038^{***}		0.060^{***}
			(0.004)		(0.006)
Other Controls					
Birth order				1.453^{***}	1.424^{***}
				(0.032)	(0.032)
Religion				0.036	0.045
				(0.126)	(0.129)
Observations	37,282	37,282	36,963	37,251	36,932
Pseudo R^2	0.177	0.245	0.196	0.425	0.433

Notes: Logit coefficients. All specs include area unit and birth year FE. Probability weights; SEs clustered at area unit level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

5.2 Logit Model: Average Marginal Effects

Table 17 reports the average marginal effects (AMEs) from the logit model to facilitate direct comparison with the LPM estimates in Table 4. The AMEs capture the average change in the probability of teen pregnancy associated with a unit change in each explanatory variable.

Table 17: Average Marginal Effects: Teen Pregnancy (Logit)

	(1) Baseline	(2) + Respondent	(3) + Husband	(4) + Birth Order	(5) Full Model
Treatment Variables					
past_group	-0.200^{***}	-0.046^{***}	-0.219^{***}	-0.107^{***}	-0.103^{***}
	(0.014)	(0.014)	(0.014)	(0.011)	(0.011)
TC_noborder	0.132^{***}	0.061^{***}	0.132^{***}	0.060^{***}	0.059^{***}
	(0.012)	(0.010)	(0.013)	(0.008)	(0.008)
Wealth Controls (ref: Richest)					
Poorest	0.354^{***}	0.140^{***}	0.275^{***}	0.023^{*}	0.025^{**}
	(0.014)	(0.014)	(0.015)	(0.012)	(0.012)
Poorer	0.296^{***}	0.106^{***}	0.228^{***}	0.025^{**}	0.025^{**}
	(0.014)	(0.014)	(0.015)	(0.011)	(0.012)
Middle	0.229^{***}	0.085^{***}	0.175^{***}	0.021^{*}	0.020^{*}
	(0.014)	(0.014)	(0.015)	(0.011)	(0.011)
Richer	0.133^{***}	0.042^{***}	0.093^{***}	0.008	0.006
	(0.014)	(0.013)	(0.014)	(0.010)	(0.011)
Respondent Characteristics					
Resp. work	-0.061^{***}	-0.004	-0.041^{***}	0.001	0.002
	(0.010)	(0.009)	(0.010)	(0.008)	(0.008)
No education		0.512^{***}		0.306^{***}	0.279^{***}
		(0.021)		(0.017)	(0.018)
Primary edu.		0.522^{***}		0.322^{***}	0.294^{***}
		(0.022)		(0.017)	(0.018)
Secondary edu.		0.384^{***}		0.242^{***}	0.222^{***}
		(0.020)		(0.016)	(0.016)
Resp. age		-0.027^{***}		-0.053^{***}	-0.057^{***}
		(0.001)		(0.001)	(0.001)
Husband Characteristics					
Husband work			0.032		0.011
			(0.027)		(0.022)
Husb. no edu.			0.211^{***}		0.037^{***}
			(0.016)		(0.013)
Husb. primary			0.206^{***}		0.043^{***}
			(0.014)		(0.012)
Husb. secondary			0.146^{***}		0.031^{***}
			(0.012)		(0.010)
Husband age			0.007^{***}		0.007^{***}
			(0.001)		(0.001)
Other Controls					
Birth order				0.172^{***}	0.166^{***}
				(0.002)	(0.002)
Religion				0.004	0.005
				(0.015)	(0.015)
Observations	37,282	37,282	36,963	37,251	36,932
Pseudo R^2	0.177	0.245	0.196	0.425	0.433

Notes: AMEs from logit model. All specs include area unit and birth year FE. Probability weights; SEs clustered at area unit level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The AMEs are remarkably close to the LPM estimates, confirming that the linear approximation performs well in this setting. In the full model (Specification 5), the AME of -0.103 for `past_group` from the logit compares closely with -0.095 from the LPM—

both point to roughly a 10 percentage point reduction in teen pregnancy attributable to Arab Spring exposure. Similarly tight correspondence holds for `TC_noborder` (0.059 vs. 0.067), and for the coefficients on birth order, education, and respondent age. The pseudo R^2 of 0.433 indicates strong model fit, comparable to the LPM's adjusted R^2 of 0.401. This consistency across linear and nonlinear specifications strengthens our confidence in the estimated treatment effects.

6 Summary of Findings and Discussion

Table 18 consolidates the key findings across all analyses.

Table 18: Summary of Key Findings

Analysis	Key Result	past_group	Significant?
Main Estimation (All Women, $N \approx 37{,}762$)			
Teen pregnancy	Reduced by 9.5 pp	-0.095	***
Age first birth	Increased by 1.1 years	1.125	***
Age first marriage	Increased by 1.0 year	0.973	***
Unskilled manual	Reduced by 17.2 pp	-0.172	***
Schooling (overall)	No significant effects	—	No
Heterogeneity (Teen Sample, $N = 826$)			
Unskilled manual	Larger effect (-39.2 pp)	-0.392	***
Marriage age	Not significant for teens	0.444	No
Secondary edu.	Marginal positive effect	0.308	*
Mechanism			
Marriage age	Delayed marriage (+1.0 year)	0.973	***
Spousal age gap	Reduced by 2.0 years (30%)	-2.017	***
Husb. education	No partner selection shift	—	No
Women's employment	No LFP channel	0.004	No
Schooling	No education channel	—	No
Occupation	Reduced unskilled labor	-0.172	***
Alternative Explanations			
Contraceptive use	No treatment effect	0.012	No
Modern methods	No treatment effect	-0.005	No
Miscarriage (all)	No effect (placebo passes)	-0.013	No
Miscarriage (teen)	No effect (placebo passes)	-0.040	No
Logit Model (Nonlinear Specification, $N \approx 36{,}932$)			
Teen preg. (coefficient)	Large negative log odds	-0.887	***
Teen preg. (AME)	Reduced by 10.3 pp	-0.103	***
Pseudo R^2	0.433 (comparable to LPM Adj. $R^2 = 0.401$)		

Notes: All results from the full model (Specification 5). "pp" denotes percentage points. AME = average marginal effect from logit. *** $p < 0.01$, * $p < 0.10$.

The main takeaways from the analysis are as follows.

1. **The Arab Spring reduced teen pregnancy.** Our estimates indicate a statistically and economically significant 9.5 percentage point reduction in teen pregnancy

attributable to exposure to the uprising. This effect is robust across specifications and survives the inclusion of a comprehensive set of controls.

2. **Two primary mechanisms emerge: delayed marriage and changed marriage matching.** The results show that Arab Spring exposure increased the age of first marriage by approximately one year and reduced the spousal age gap by 2 years—a 30% reduction from the mean of 6.65 years. The narrowing of the age gap suggests that the uprising disrupted traditional marriage norms, leading to more egalitarian partnerships in which women exercised greater agency over reproductive timing. Importantly, these effects did *not* operate through changes in women's or husbands' education, nor through shifts in women's labor force participation. The mechanism appears to be fundamentally about *how* and *when* women married.

3. **A significant shift in occupational composition is evident.** The uprising reduced women's participation in unskilled manual labor by 17.2 percentage points, with an even larger effect among teens (-39.2 pp). However, overall employment rates did not change, indicating a reallocation across *types* of work rather than a change in the *level* of labor force participation.

4. **The treatment effects are heterogeneous by age.** The estimates reveal qualitatively different patterns for teens compared with all women: occupational effects are amplified, marriage age effects disappear, and marginal educational gains emerge. These differences suggest that the Arab Spring affected younger and older women through distinct channels.

5. **The results are robust to nonlinear specification.** A logit model for the binary teen pregnancy outcome yields average marginal effects that closely match the LPM estimates. The logit AME of `past_group` is -0.103 (vs. -0.095 in the LPM), and the pseudo R^2 of 0.433 is comparable to the LPM adjusted R^2 of 0.401. The consistency across linear and nonlinear models reinforces confidence in our results.

6. **The placebo test is passed.** Using miscarriage probability as a placebo outcome, no treatment effect is found in either sample, confirming that our findings reflect genuine behavioral changes in fertility decisions rather than differential changes in pregnancy outcomes.

Data: Egypt DHS 2005, 2008, 2014 (combined $N \approx 38{,}000$).

Complete results: Available in `Results/summary/` directory (formatted Excel workbooks)

and individual CSV/PNG files in `Results/{main_estimation, hetro_analysis, mechanism, robustness_checks}/{Pallab, Yeganeh}/`.